

HDOC DAY – 14

QUESTION:

- Check whether all its digits are unique.
 - **E.g. 1:** Number: 12345, Output: Unique
 - **E.g. 2:** Number: 12234, Output: Not Unique
- Find the frequency of each digit from `0` to `9`.
 - **E.g.** Number: 112333, freq_0 = 0, freq_1 = 2, freq_2 = 1, freq_3 = 3, freq_4 = 0, freq_5 = 0, freq_6 = 0, freq_7 = 0, freq_8 = 0, freq_9 = 0

ALGORITHM:

1. Start.
2. Display the menu: unique-digit check, digit frequency, and exit.
3. Read the user's choice.
4. Use switch-case on the choice.

Case 1:

- a. Read positive integer n.
- b. Create seen[10] and initialize it to zero.
- c. Extract each digit using $n \% 10$. If the digit is already in seen, print "Not Unique"; otherwise mark it in seen.
- d. If no digit repeats, print "Unique".

Case 2:

- a. Read positive integer n.
- b. Create frequency[10] and initialize it to zero.
- c. Extract each digit using $n \% 10$ and increment frequency[digit].
- d. Print frequency[0] to frequency[9].

Case 3: Exit the program.

Default: Display invalid-choice message.

5. Repeat until choice 3 is selected.

6. Stop.

C program:

```
#include <stdio.h>
```

```
/* Returns 1 when every digit of n occurs exactly once, otherwise 0. */
```

```
int hasUniqueDigits(long long n)
```

```
{
```

```
    int seen[10] = {0};
```

```
    while (n > 0) {
```

```
        int digit = (int)(n % 10);
```

```
        if (seen[digit] == 1) {
```

```
            return 0;
```

```
        }
```

```
        seen[digit] = 1;
```

```
        n /= 10;
```

```
    }
```

```
    return 1;
```

```
}
```

```
void printDigitFrequencies(long long n)
```

```
{
```

```
    int frequency[10] = {0};
```

```
    int digit;
```

```
    while (n > 0) {
```

```
        digit = (int)(n % 10);
```

```
        frequency[digit]++;
```

```

        n /= 10;
    }
    printf("Digit frequencies:\n");
    for (digit = 0; digit <= 9; digit++) {
        printf("freq_ %d = %d\n", digit, frequency[digit]);
    }
}

int main(void)
{
    int choice;
    long long number;

    do {
        printf("\n--- Digit Operations Menu ---\n");
        printf("1. Check whether all digits are unique\n");
        printf("2. Find frequency of each digit (0-9)\n");
        printf("3. Exit\n");
        printf("Enter your choice: ");

        if (scanf("%d", &choice) != 1) {
            printf("Invalid menu input. Program terminated.\n");
            return 1;
        }

        switch (choice) {
        case 1:
            printf("Enter a positive integer: ");

```

```
if (scanf("%lld", &number) != 1 || number <= 0) {  
    printf("Invalid input. Please enter a positive integer.\n");  
} else if (hasUniqueDigits(number)) {  
    printf("Output: Unique\n");  
} else {  
    printf("Output: Not Unique\n");  
}  
break;
```

case 2:

```
printf("Enter a positive integer: ");  
if (scanf("%lld", &number) != 1 || number <= 0) {  
    printf("Invalid input. Please enter a positive integer.\n");  
} else {  
    printDigitFrequencies(number);  
}  
break;
```

case 3:

```
printf("Program exited successfully.\n");  
break;
```

default:

```
printf("Invalid choice. Please select 1, 2, or 3.\n");  
}  
} while (choice != 3);
```

```
return 0;
```

}

```
int main()
{
    int choice;
    long long number;

    do {
        printf("\n — MENU — \n");
        printf("1. Check unique digits\n");
        printf("2. Find frequency of digits\n");
        printf("3. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                printf("Enter a positive integer: ");
                scanf("%lld", &number);

                if (number <= 0) {
                    printf("Invalid input. Enter a positive integer.\n");
                } else if (hasUniqueDigits(number)) {
                    printf("Output: Unique\n");
                } else {
                    printf("Output: Not Unique\n");
                }
                break;

            case 2:
                printf("Enter a positive integer: ");
                scanf("%lld", &number);

                if (number <= 0) {
                    printf("Invalid input. Enter a positive integer.\n");
                } else {
                    printDigitFrequencies(number);
                }
                break;

            case 3:
                printf("Program exited successfully.\n");
                break;

            default:
                printf("Invalid choice. Enter 1, 2, or 3.\n");
        }
    } while (choice != 3);

    return 0;
}
```

Test table:

Test case	Choice	number	Expected result
1	1	12345	Output: unique
2	1	12234	Output: not unique
3	2	112333	Freq_0=0, freq_1=2, freq_2=1,

			freq_3=3; freq_4 to freq_9=0
4	4	--	Invalid choice message
5	1	0	Invalid positive integer message

```

— MENU —
1. Check unique digits
2. Find frequency of digits
3. Exit
Enter your choice: 1
Enter a positive integer: 12345
Output: Unique

— MENU —
1. Check unique digits
2. Find frequency of digits
3. Exit
Enter your choice: 1
Enter a positive integer: 12234
Output: Not Unique

— MENU —
1. Check unique digits
2. Find frequency of digits
3. Exit
Enter your choice: 1
Enter a positive integer: 101
Output: Not Unique

— MENU —
1. Check unique digits
2. Find frequency of digits
3. Exit
Enter your choice: 112333
Invalid choice. Enter 1, 2, or 3.

— MENU —
1. Check unique digits
2. Find frequency of digits
3. Exit
Enter your choice: 2
Enter a positive integer: 112333

Digit Frequencies:
freq_0 = 0
freq_1 = 2
freq_2 = 1
freq_3 = 3
freq_4 = 0
freq_5 = 0
freq_6 = 0
freq_7 = 0
freq_8 = 0
freq_9 = 0

```